SHIFTING TORNADO HAZARD IN THE UNITED STATES AND EMERGING INSURANCE EXPOSURE

Stakeholder:

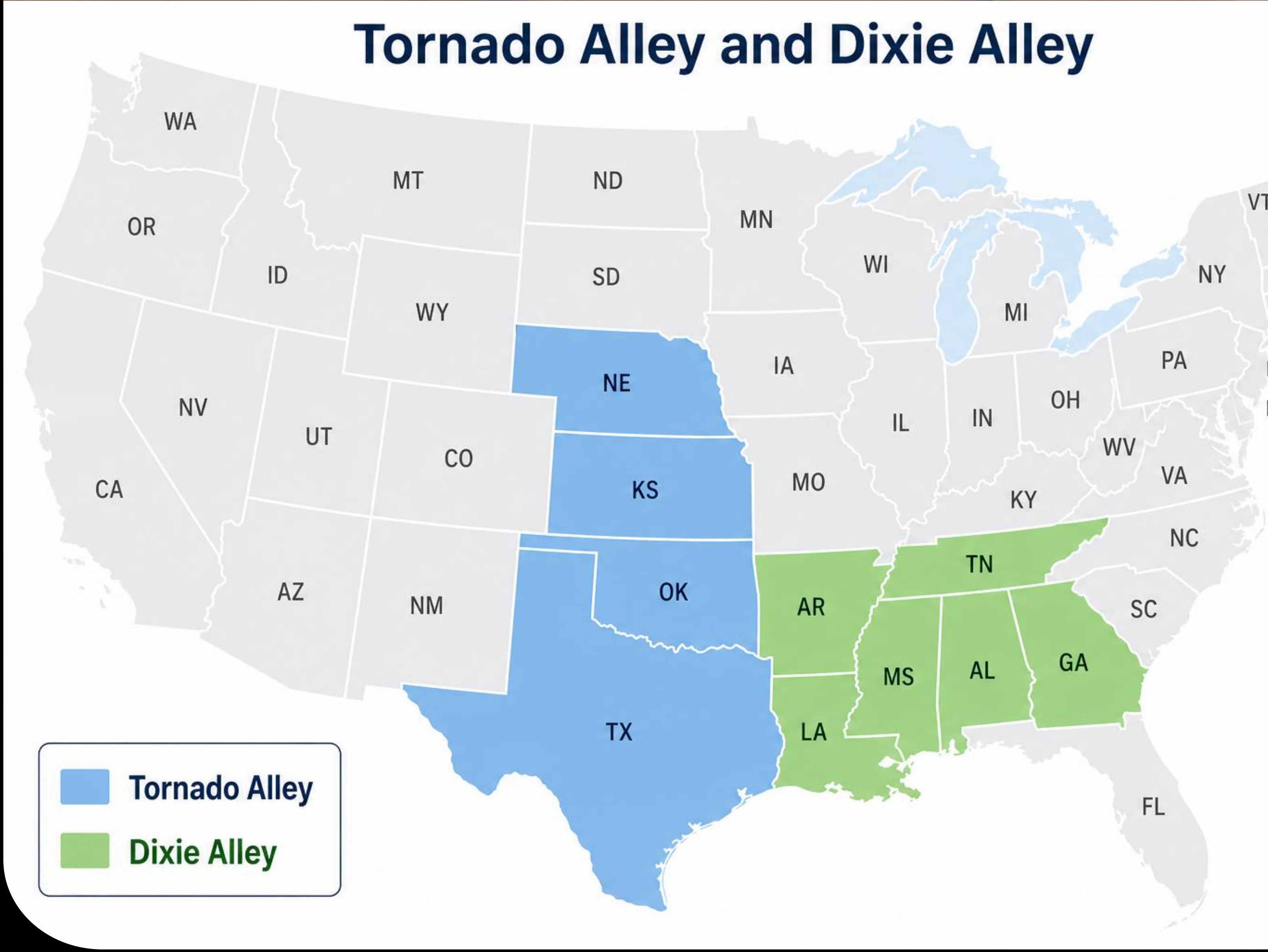
Insurance Company ABC

Property Risk Analysts and Underwriting Team

Team Gamma
Presenter: Jillian Baker

BACKGROUND

- Recent research indicates that tornado activity may be shifting geographically from Tornado Alley towards Dixie Alley
- Extends prior research by linking shifting tornado hazard to present-day insurance exposure at the county-level
- Potentially higher risk of insurance exposure:
  - Population
  - Housing Units
  - Median Home Value
- Analysis into which counties are becoming more hazardous over time



Reference: Gensini & Brooks (Spatial Trends in United States Tornado Frequency, 2018) found significant eastward shifts in tornado frequency supporting increasing trends in the Southeast's Dixie Alley and decreasing trends in the Great Plains Tornado Alley.

BACKGROUND

RESEARCH QUESTION & HYPOTHESIS

- **Research Question:**
  - Is tornado hazard shifting towards Dixie Alley?
  - Do high-risk counties also have a higher insurance exposure?
- **Hypothesis:** Tornado hazard is shifting from the Tornado Alley region towards Dixie Alley due to increasing tornado frequency and severity in the Southeast over time

PREDICTION

- Counties located in Dixie Alley will be more likely classified as high tornado hazard in the following year compared to Tornado Alley and will exhibit a higher present-day insurance risk

METHODS: DATA DETAILS

DATA SOURCE

- **National Oceanic and Atmospheric Administration (NOAA)** Severe Weather Database for all historical tornado data (1950-2024)
  - Detailed tornado records: Location, intensity, property loss
- **American Community Survey** for all county-level exposure data (tidycensus in R) for 5-year estimates for the variable 2024:
  - Estimate Total Population
  - Estimate Total Housing Units
  - Estimate Median Home Value (dollars)

- Time range: 1979-2024
- Observations: 31,804
- Unit of analysis: Aggregated by county-year
- Features (Predictor Variables): 6
  - Exposure (Insurance Risk)
    - Population Estimate
    - Housing Unit Estimate
    - Median Value Estimate
  - Hazard Characteristics
    - Average Length (tornado path length in miles)
    - Average Width (tornado path width in yards)
  - Geographic Location
    - Region Category (Tornado Alley vs. Dixie Alley)

METHODS: DATA DETAILS

DATA CLEANING

- Datasets merged using County FIPS identifiers
- Retained observations between 1979-2024
- Tornado events aggregated at a county-level
- Missing/invalid variables addressed
- F/EF Scale standardized to 0-5 scale
- Property Loss values cleaned due to inconsistent reporting

RESPONSE/TARGET VARIABLE

- Binary target variable for prediction of high tornado hazard next year
- Composite score:
  - Tornado Frequency
  - Tornado intensity
  - Strong tornado count (F/EF 3 or greater)
  - Property Loss
- Standardized using z-scores to ensure comparability across differing scales
- Shifted forward one year to look at future hazard conditions
- Composite scores at or above the 65th percentile:
  - Flagged as 1 = high hazard next year
  - Otherwise, 0 = not high hazard next year

MODELING APPROACH

MODELS USED

- 3 models used:
  - Logistic Regression (baseline mode)
  - random forest
  - XGBoost
- Time-based train/test split
  - Training Set: 1979-2014
  - Testing Set: 2015-2024
    - Model evaluated on future observations

HYPERPARAMETER TUNING, CROSS VALIDATION AND EVALUATION

- Applied to random forest and XGBoost models
  - Leaving baseline model (Logistic Regression) untuned
- Randomized search
- Cross validation
  - Stratified k-fold
  - Repeated k-fold
- Recall: Primary evaluation method. Essential to accurately identify high-risk counties.
- PR-AUC: How well each model performed in determining risk
- Confusion matrix and sensitivity of each model

Model	Recall	PR-AUC	Interpretation
Logistic Regression	94.20%	0.6861	Baseline model with no tuning. Over predicts false positive county_year observations as high risk
random forest	51.49%	0.6745	Misses too many high-risk county_year observations
XGBoost	91.15%	0.7075	Best performing model. Captures most accurate high-risk county_year observations and fewer missed.

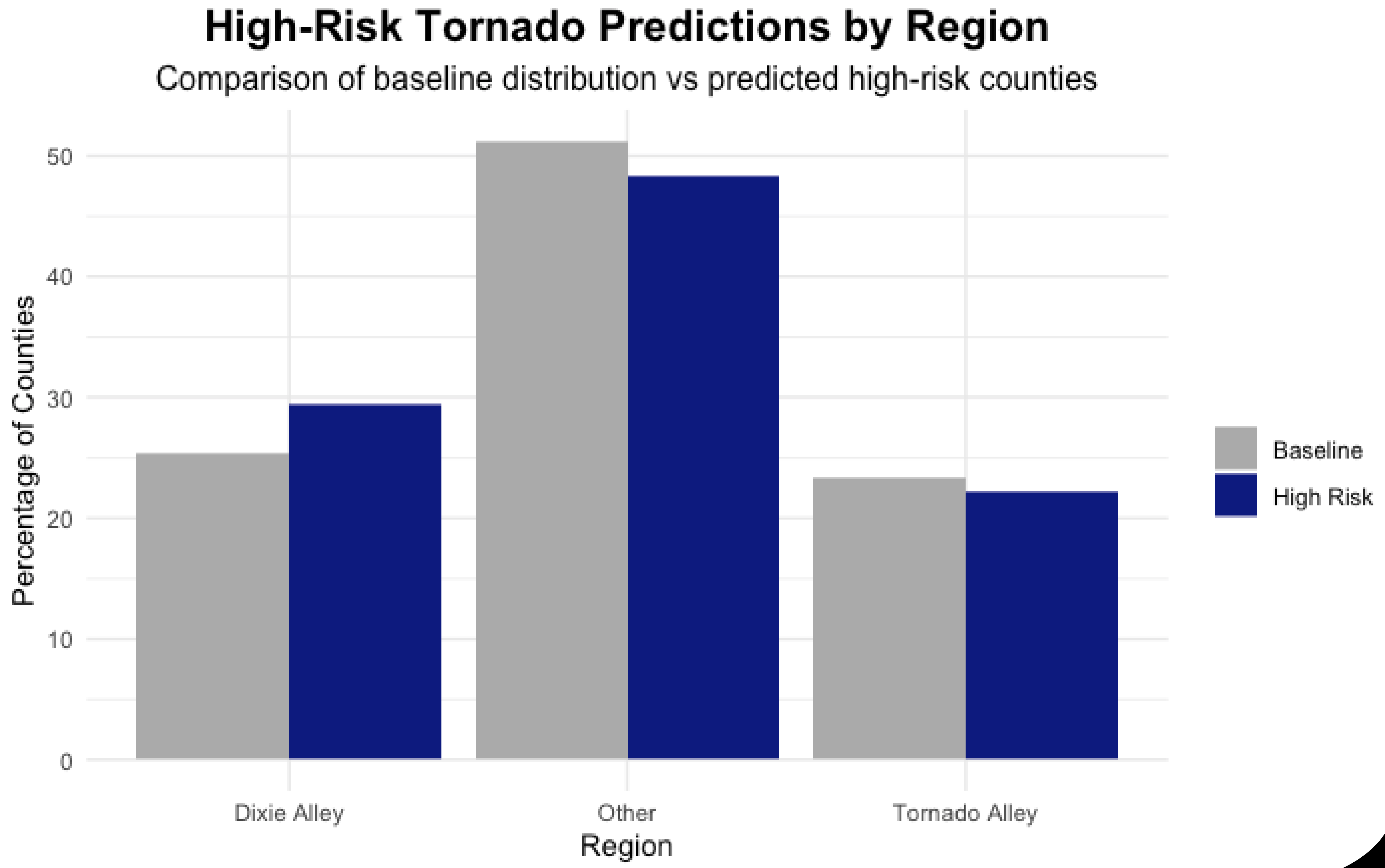
Figure 2: Model comparison of results for stratified k-fold cross validation (k=5)

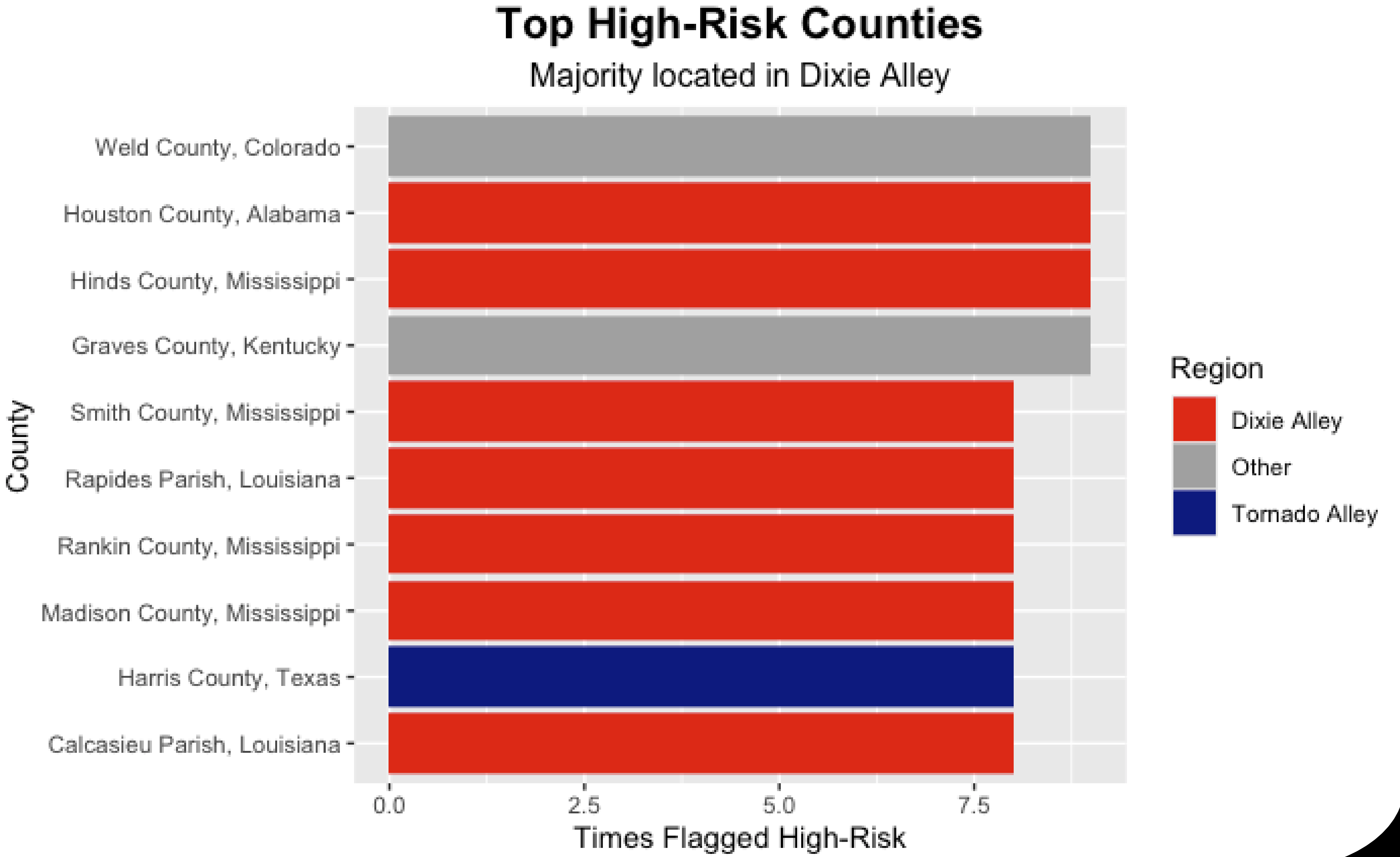
BEST PERFORMING MODEL: XGBOOST

Provided best balance with high recall sensitivity (91.15%) and PR-AUC (0.7075)

Supported by confusion matrix achieving a strong number of true positives and lower false positives

Indicating it performs well at accurately predicting high-risk counties with fewer misses.

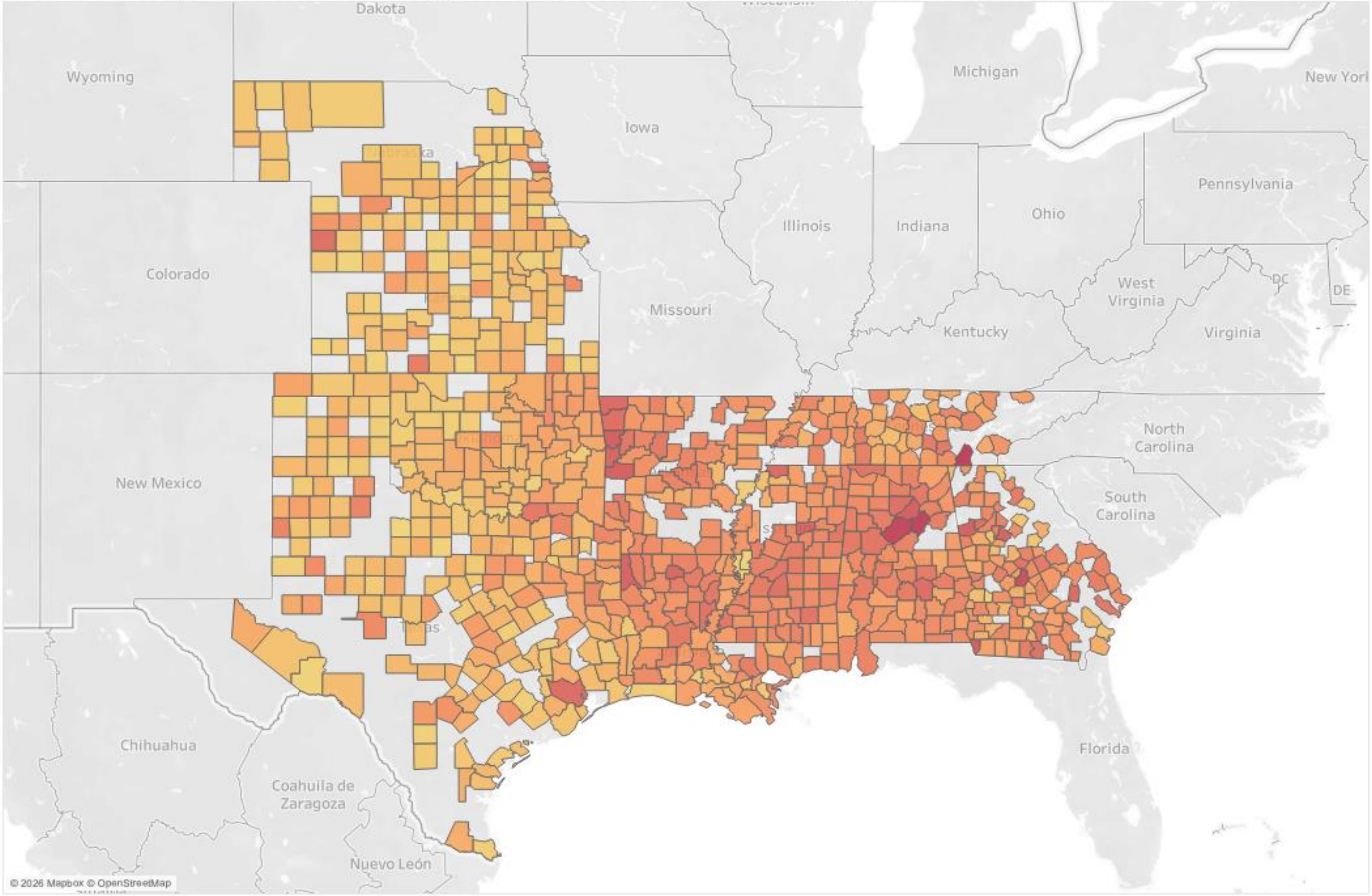

High-Risk Tornado Predictions by Region
Comparison of baseline distribution vs predicted high-risk counties
Percentage of Counties
0
10
20
30
40
50
Dixie Alley
Other
Tornado Alley
Region
Baseline
High Risk

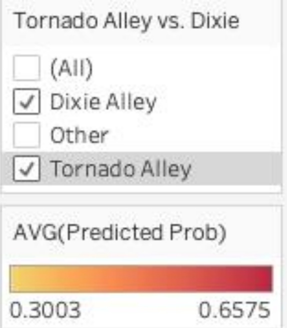


Top High-Risk Counties
Majority located in Dixie Alley
Weld County, Colorado
Houston County, Alabama
Hinds County, Mississippi
Graves County, Kentucky
Smith County, Mississippi
Rapides Parish, Louisiana
Rankin County, Mississippi
Madison County, Mississippi
Harris County, Texas
Calcasieu Parish, Louisiana
County
0.0
2.5
5.0
7.5
Times Flagged High-Risk
Region
Dixie Alley
Other
Tornado Alley

Predicted Tornado Hazard Probability for High-Risk Counties

Includes only counties that are flagged as high-risk with darker shading indicating a higher predicted probabilty





LIMITATIONS AND NEXT STEPS

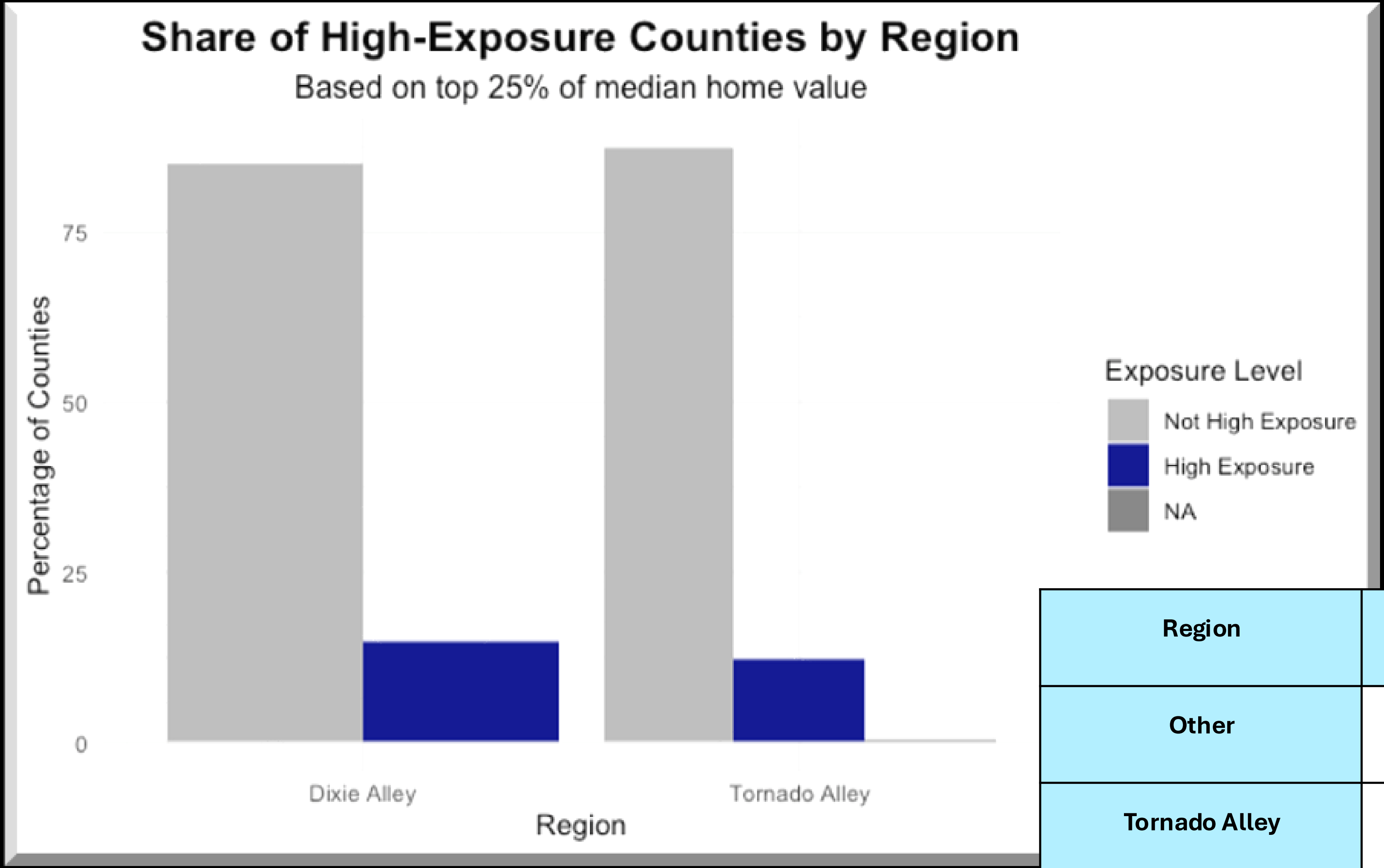
LIMITATIONS AND CONSIDERATIONS

- Model relies historical tornado data and may not reflect future climate shifts
- Spatial dependence likely violated
- Exposure is measured in population and home value and not actual insurance losses
- Multicollinearity in exposure variables, however, retained for relevance

NEXT STEPS

- To be used as a decision-support tool
- Could look to incorporate spatial modeling techniques
- Integrate actual insurance loss data

INSURANCE EXPOSURE IN DIXIE ALLEY



Region	Average Population	Average Home Value
Other	171,065	218,289
Tornado Alley	151,036	176,371
Dixie Alley	80,331	170,448

CONCLUSION

SUMMARIZED FINDINGS

- Tornado hazard is shifting towards Dixie Alley
- High-risk counties are increasingly concentrated in the Southeast
- Dixie Alley has a higher proportion of high-exposure counties despite lower averages
- Dixie Alley represents a new hotspot for tornado hazard and insurance risk
    - Potential need for underwriting and pricing strategies by Insurance Company ABC